

Design and optimization of dual-band sonic topological insulators based on genetic algorithm

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ABSTRACT

The 2D sonic topological insulators (STIs), which can support topological edge states and corner states, provide a platform for acoustic manipulation. We utilize genetic algorithm (GA) to optimize the C_{4v} symmetric square lattices to achieve dual-band STIs, and the center frequency of the second topological bandgap is twice that of the first one. The topological properties of the optimized lattice are investigated based on the parities at the high symmetry points. In a further way, dual-band topological edge states and double corner states are verified by finite element method simulation in the domain wall and bending structures, respectively. A GA optimization is also used for optimizing the nearest two unit cells at the domain wall, and the bandwidths of edge states are widened. Our work provides an efficient way for achieving the frequency-doubled topological acoustic wave manipulation and multiband acoustic topological cavities.

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I. INTRODUCTION

Phononic crystals (PCs) are a class of novel artificial materials with periodic structures and have been extensively investigated for the manipulation of acoustic and elastic waves.^{1–5} In recent years, based on the study of topological properties in electronic and photonic systems, many researchers have extended the topological theory to the field of acoustics.^{6,7} Because of the topological protection of sonic topological insulators (STIs), the edge states exhibit extraordinary robustness against disorders, which provides a new approach for wave manipulation with immunity to defects.^{8–14} Conventional STIs adhere to the bulk-boundary correspondence principle, and the topological invariants of bulk band structures determine the existence of edge states.^{15–19} Currently, various STIs with quantum valley Hall effect (QVHE),²⁰ quantum spin Hall effect (QSHE),²¹ as well as Chern insulators²² have been realized.

Compared to conventional STIs, higher-order sonic topological insulators (HSTIs) exhibit novel topological properties.^{23–25} The topological states of HSTIs emerge at lower-dimensional boundaries or corners than conventional STIs.^{26–30} For example, 2D second-order STIs have 1D edge states and 0D corner states. These lower-dimensional corner and edge states are topologically protected and are stable as long as the system's symmetry is

preserved.³¹ In addition, based on the Su–Schrieffer–Heeger (SSH) model, HSTIs with various symmetries have been reported, including C_{3v} symmetry, C_{4v} symmetry, C_{6v} symmetry, and glide symmetry, as well as novel STIs combining different symmetries.^{32–38}

To realize the topological states, it is necessary to open the omnidirectional bandgap of PCs. Compared to traditional trial-and-error methods, employing various algorithms to optimize PCs can significantly improve efficiency and reduce time costs of research.^{39–41} Some teams have successfully realized novel STIs with multifunctional properties through topological optimization,⁴² machine learning,⁴³ and inverse design.⁴⁴ In addition, researchers have optimized the structural configurations and material properties^{45–48} of PCs to achieve distinct optimization objective, including multiple topological bandgaps,⁴⁹ maximizing the relative bandgap width⁵⁰ and double bands of HSTIs.⁵¹ However, there are few reports on the design and optimization of dual-band STIs with double frequency.

In this work, we develop a design method based on genetic algorithm (GA) to achieve the dual-band sonic topological insulators (STIs) based on C_{4v} symmetric square lattices. In our proposed structure, one unit cell (UC) consists of an odd number of cylindrical scatterers, which are made of epoxy resin. We utilize

the GA with the calculation of band diagrams based on the finite element method (FEM) and obtain the optimized parameters of C_{4v} symmetric square lattices. Double bandgaps can be achieved, and the center frequency of the second bandgap is twice that of the first one. Based on the parity properties of eigenfields, the topological properties of two kinds of UCs are analyzed. By constructing the domain wall and bending structures composed of UCs with opposite topological properties, dual-band topological edge states and double corner states can be realized, respectively. Furthermore, we conduct a second-stage optimization for the nearest two UCs at the domain wall to broaden the bandwidths of edge states based on GA. Therefore, our work provides an effective method to design dual-band STIs, and find applications in nonlinear acoustic wave processing, acoustic cavities, and so on.

II. DESIGN PROCESS AND OPTIMIZATION METHOD

We design C_{4v} symmetric square lattices based on the GA optimization method for achieving dual-band two-dimensional STIs, and one unit cell consists of an odd number of cylindrical scatterers. The cylinders are made of epoxy resin with density $\rho_0 (=1180 \text{ kg/m}^3)$, embedded in an air background with density $\rho (=1.21 \text{ kg/m}^3)$. The speeds of sound in epoxy resin and air are $c (=2720 \text{ m/s})$ and $c_{\text{air}} (=343 \text{ m/s})$, respectively. For example, the STI with one unit cell containing 17 cylinders is shown in Fig. 1(a), and the two choices of unit cell are indicated by UC1A and UC1B. Considering C_{4v} symmetry, we only consider the one eighth unit cell containing four cylinders $c_i (i = 1, 2, 3, 4)$. Here, the cylinder c_2 is located on the diagonal and c_4 is located at the corner of square. The geometrical parameter space for optimization involves 12 parameters $(r_1, x_1, y_1, r_2, x_2, y_2, r_3, x_3, y_3, r_4, x_4, y_4)$, where x_i, y_i , and r_i represent the x -coordinate, y -coordinate, and radius of the i th ($i = 1, 2, 3, \text{ and } 4$) epoxy resin cylinder, respectively.

Our optimization aims to open two bandgaps simultaneously, where the center frequency of the second bandgap is twice that of the first bandgap. In a further way, we discuss the topological properties and construct edge and corner states. Here, we utilize the GA with the calculation of band diagrams based on the finite element method (FEM). The optimization process is completed on MATLAB 2023b running on a single central processing unit core (AMD Ryzen 7 5700G). COMSOL with MATLAB facilitates the programming of fitness evaluation, enabling flexible application of GA with different optimization objectives. The process and pseudocode of GA are detailed in Appendix A, and the fitness function for opening two bandgaps simultaneously is provided in Appendix B.

III. DUAL-BAND STIs BASED ON C_{4v} SYMMETRIC SQUARE LATTICE

In this section, we design and optimize STIs, based on C_{4v} symmetric square lattice, and STIs host both dual-band edge states and dual corner states.

A. Optimized band structures and topological properties

We set opening two bandgaps as the optimization goal, and the center frequency of the second bandgap is twice that of the first

bandgap. By conducting the GA based on Appendixes A and B, the optimized parameters of C_{4v} symmetric square lattice with one unit cell containing 17 cylinders are obtained (as shown in UC1 of Table I in Appendix C), and the optimized band diagram of UC1 is shown in Fig. 1(b). As can be seen, two bandgaps are simultaneously opened in the band diagram of UC1, and all calculated eigen frequencies are normalized by $2\pi c_{\text{air}}/a$. The first bandgap between the third and fourth bands ranges from $0.637(2\pi c_{\text{air}}/a)$ to $0.686(2\pi c_{\text{air}}/a)$ with the center frequency of $0.661(2\pi c_{\text{air}}/a)$, and the second bandgap between the ninth and tenth bands ranges from $1.292(2\pi c_{\text{air}}/a)$ to $1.355(2\pi c_{\text{air}}/a)$ with the center frequency of $1.323(2\pi c_{\text{air}}/a)$. It can be observed that the center frequency of the second bandgap is twice that of the first bandgap. The bandgaps originate from Bragg scattering, as evidenced by the clear $\propto 1/a$ scaling observed when varying the lattice constant. Furthermore, Fig. 10(a) in Appendix G discusses the robustness of this relationship under varying geometric parameters. Figure 1(c) shows the evolutionary history of the GA optimization for UC1. The four insets in Fig. 1(c) represent the band diagrams of UC1 during the iteration process. From left to right, they correspond to the optimization results of generations 3, 31, 77, and 141. It can be observed that no bandgap opens below generation 31. At the 77th generation, both bandgaps open simultaneously. Finally, the fitness value stabilizes at the 141st generation, and thus, we set a maximum iteration count of 200 to obtain the final optimized result.

Here, the two choices of unit cell are indicated by UC1A and UC1B, and UC1A(UC1B) can be obtained by moving UC1B(UC1A) along the diagonal direction by $(a/2, a/2)$, as shown in Fig. 1(a). For demonstrating the topological properties, Fig. 1(d) shows the eigen modes of the different bands of UC1A and UC1B at the high-symmetry points Γ and X_i , where unipolar or quadrupolar modes indicate even (+) parity, while dipole modes correspond to odd (−) parity. Previous studies have shown that the 2D polarization $\mathbf{P} = (P_x, P_y)$ can be used to characterize the topological properties of C_{4v} symmetric structures.^{52–54} It is calculated as follows:

$$P_i = \frac{1}{2} \left(\sum_n q_i^n \bmod 2 \right) (-1)^{q_i^n} = \frac{\eta_n(X_i)}{\eta_n(\Gamma)}, \quad (1)$$

where i denotes the direction x, y ; n represents the band order; η_n indicates the parity at the high symmetry points Γ and X_i for the n th band. Additionally, $\sum q_i^n$ denotes the sum of q_i^n from 1 to n , and mod is the modulo operation. The 2D polarization (P_x, P_y) can be calculated using Eq. (1). Because of the C_{4v} symmetry of the structure, P_x is equal to P_y . Based on the parity properties of UC1A and UC1B in Fig. 1(d) and Eq. (1), the 2D polarization for the first bandgap of UC1A is $\mathbf{P} = (0, 0)$, indicating it is topologically trivial. In contrast, UC1B exhibits $\mathbf{P} = (1/2, 1/2)$ for the first bandgap, indicating it is topologically nontrivial. In the same way, the second bandgap of UC1B is topologically trivial and the second bandgap of UC1A is topologically nontrivial. The interface between UC1A and UC1B supports topological edge states in the two bandgaps due to the opposite topological properties. In addition, the topological corner charge Q_c can be expressed as $Q_c = 4P_x P_y$.^{55,56} We find the Q_c of UC1A in the first bandgap is 0, while that of UC1B is 1. In addition, the Q_c of UC1A in the

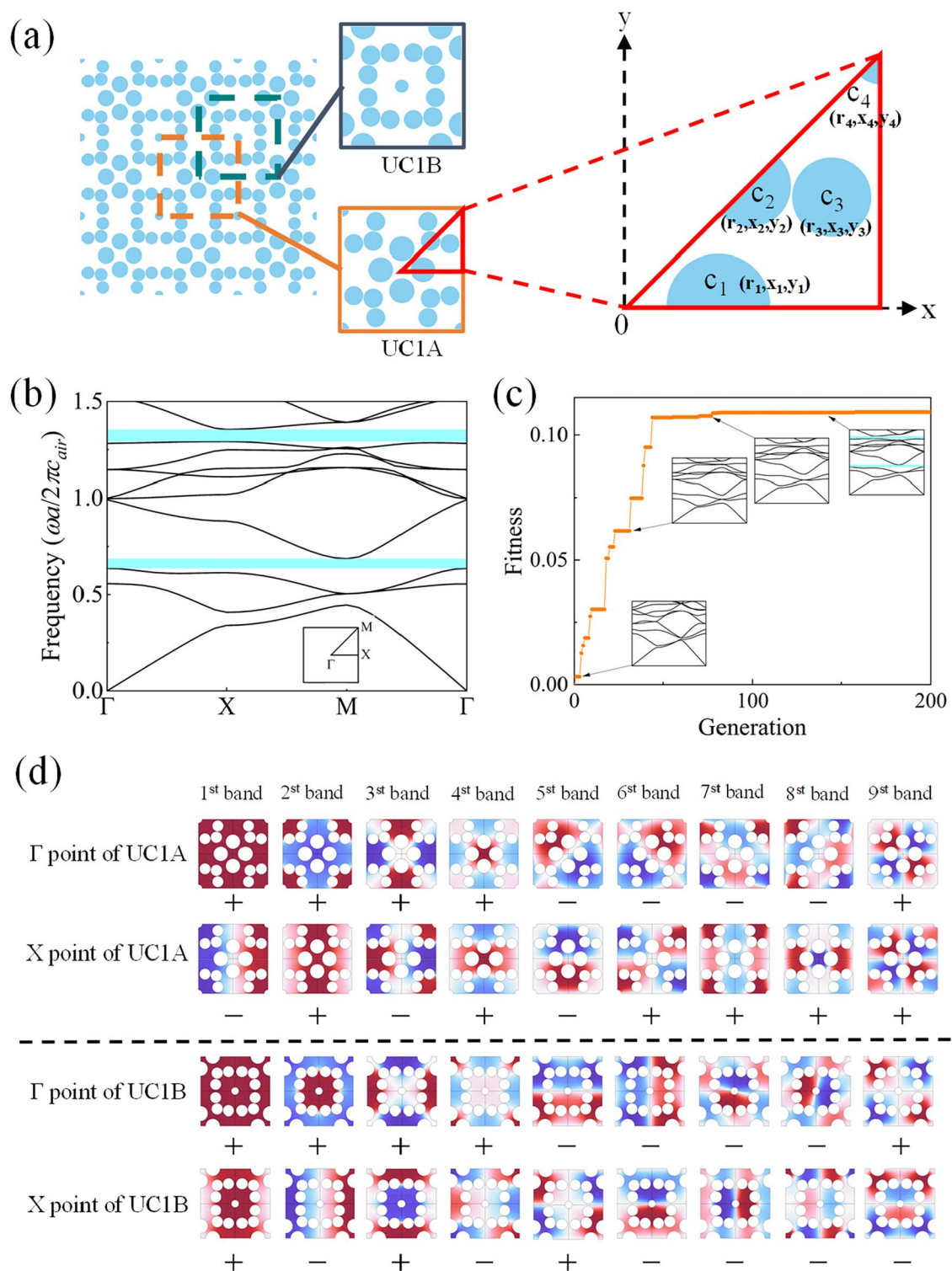


FIG. 1. (a) Configuration of UC1 with 17 cylinders in one cell and two choices of unit cell. (b) Band diagram of UC1 after 200 iterations. (c) Evolutionary history of GA optimization for UC1. (d) The parity of the nine bands of UC1A and UC1B at high-symmetry points.

second bandgap is 1, while that of UC1B is 0. UC1A and UC1B exhibit nonzero corner charges, indicating that topological corner states emerge within the first and second bandgaps.

B. Topological edge states in dual-band STIs

Next, we assemble UC1As and UC1Bs to form domain wall structures shown in Fig. 2(a), and the opposite topological properties in the two bandgaps ensure the dual-band topological edge states. Based on the FEM simulation, the projected band diagram is calculated as shown in Fig. 2(b). Here, the blue region represents bandgaps, and the gray regions represent bulk bands. It can be observed that two edge states appear within the two bandgaps, plotted with orange and red lines, respectively. We also plot the sound pressure fields as shown in Fig. 2(c), at the frequencies of $0.658(2\pi c_{\text{air}}/a)$ and $1.308(2\pi c_{\text{air}}/a)$, respectively, corresponding to the selected frequencies indicated by dots in Fig. 2(b). It can be seen that the sound pressure fields are highly localized around the interfaces between UC1A and UC1B. Therefore, dual-band topological edge states can be achieved, and the frequency multiplication relationship between two topological edge states shows the

application of nonlinear acoustic wave processing. However, it is found that the two edge states merge into bulk bands with increasing k_x , which leads to a limited working range for the edge states. The first edge state ranges from $0.635(2\pi c_{\text{air}}/a)$ to $0.666(2\pi c_{\text{air}}/a)$, and the second edge state ranges from $1.292(2\pi c_{\text{air}}/a)$ to $1.31(2\pi c_{\text{air}}/a)$. To enlarge the bandwidth of the edge states and fully utilize topological bandgaps, we perform further optimization to broaden bandwidths of the edge states, and the results are shown in Fig. 7 of Appendix D.

C. Double corner states in dual-band STIs

Due to the nonzero corner charges of UC1A and UC1B, the corner states in two bandgaps can be achieved in the bending structure, which is composed of UC1A surrounded by UC1B, as shown in Fig. 3(a). Figure 3(b) shows the eigenvalue spectrum, and blue regions represent the bandgaps. Here, gray dots represent bulk bands, and blue (red) dots represent edge (corner) states. Figure 3(c) plots the sound pressure fields for edge and corner states, which correspond to $C_1[0.689(2\pi c_{\text{air}}/a)]$, $C_2[1.352(2\pi c_{\text{air}}/a)]$, $E_1[0.656(2\pi c_{\text{air}}/a)]$, and $E_2[1.305(2\pi c_{\text{air}}/a)]$, respectively.

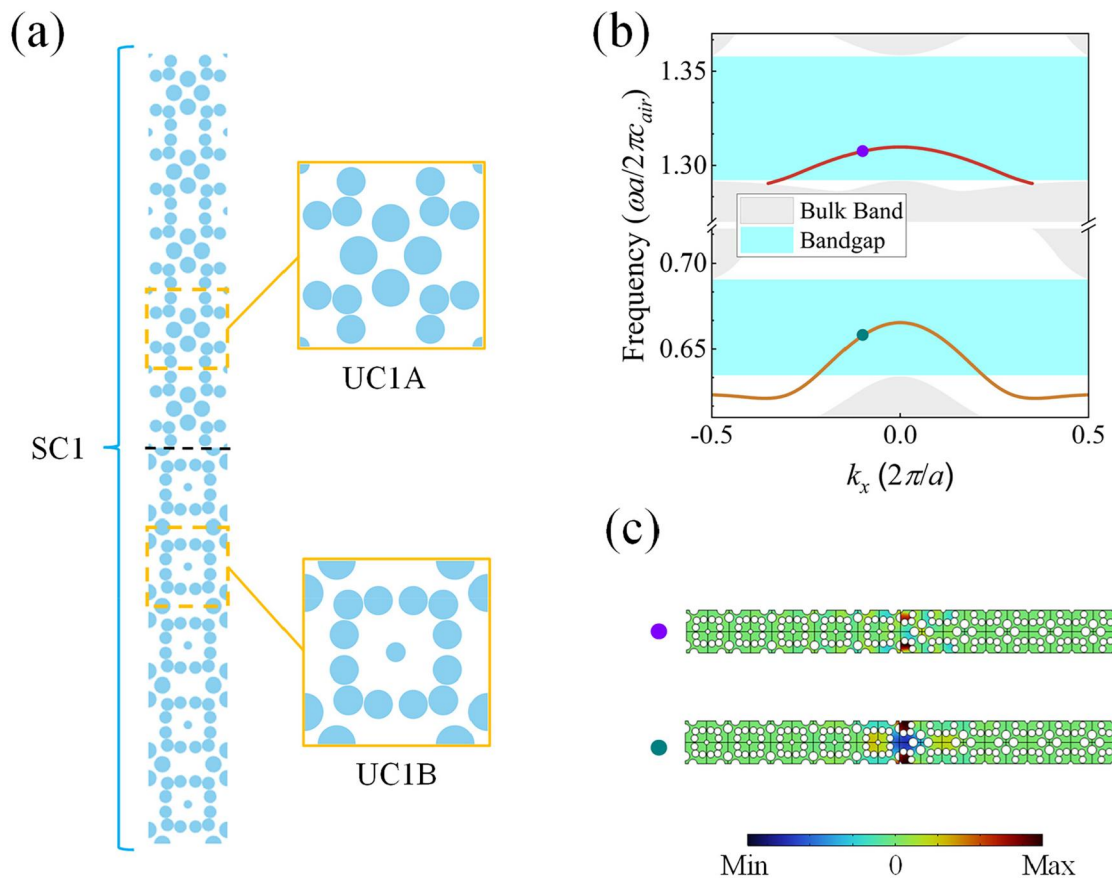


FIG. 2. (a) Domain wall composed of UC1A and UC1B; (b) projected band diagrams of the SC1; (c) the sound pressure fields of edge states in (b).

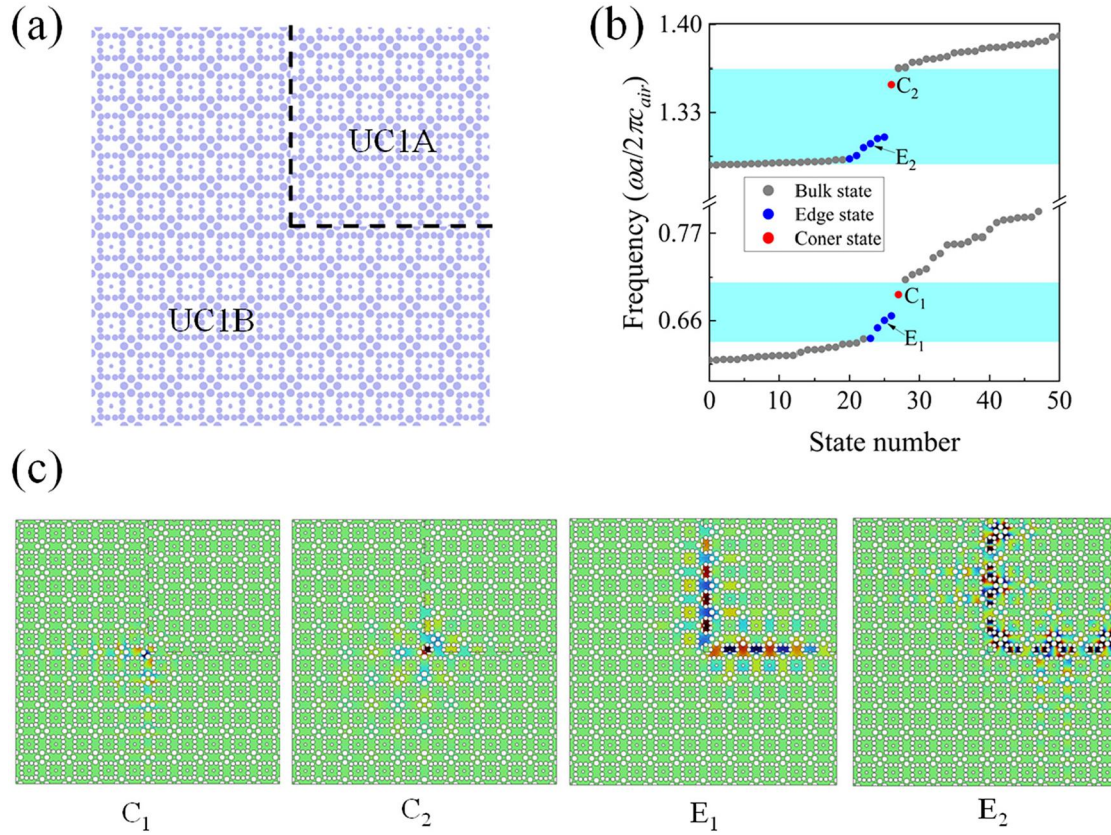


FIG. 3. (a) Bending structure composed of UC1A surrounded by UC1B; (b) Eigenvalue spectrum; (c) sound pressure fields for the corner states and edge states in (b).

Obviously, the edge states are localized along the two boundaries formed by UC1A and UC1B, and the corner states are strongly confined to the corner region. Therefore, dual-band HSTIs can be achieved based on GA.

D. Universality of the optimization method

Furthermore, to demonstrate the universality of the proposed method, we design another three C_{4v} symmetric square lattices named UC2, UC3, and UC4, and one unit cell of UC2, UC3, and UC4 have 15, 19, and 21 cylinders, respectively. With the same optimization process as UC1, we set opening two bandgaps as the optimization goal, and the center frequency of the second bandgap is twice that of the first bandgap. By conducting the GA, the optimized parameters for UC2, UC3, and UC4 are obtained (as shown in Table I of Appendix C). The optimized configurations of the three sonic crystals are shown in Figs. 4(a), 4(d), and 4(g), and one eighth unit cells are plotted in insets. The optimized band diagrams of UC2, UC3, and UC4 are shown in Figs. 4(b), 4(e), and 4(h), respectively. For UC2, the first bandgap between the fifth and sixth bands ranges from $0.853(2\pi c_{\text{air}}/a)$ to $0.940(2\pi c_{\text{air}}/a)$ with the center frequency of $0.896(2\pi c_{\text{air}}/a)$, and the second bandgap between the 13th and 14th bands ranges from $1.703(2\pi c_{\text{air}}/a)$ to $1.885(2\pi c_{\text{air}}/a)$

with the center frequency of $1.794(2\pi c_{\text{air}}/a)$. For UC3, the first bandgap between the fifth and sixth bands ranges from $0.860(2\pi c_{\text{air}}/a)$ to $0.930(2\pi c_{\text{air}}/a)$ with the center frequency of $0.895(2\pi c_{\text{air}}/a)$, and the second bandgap between the 13th and 14th bands ranges from $1.667(2\pi c_{\text{air}}/a)$ to $1.918(2\pi c_{\text{air}}/a)$ with the center frequency of $1.793(2\pi c_{\text{air}}/a)$. For UC4, the first bandgap between the seventh and eighth bands ranges from $1.120(2\pi c_{\text{air}}/a)$ to $1.206(2\pi c_{\text{air}}/a)$ with the center frequency of $1.163(2\pi c_{\text{air}}/a)$, and the second bandgap between the 21st and 22nd bands ranges from $2.273(2\pi c_{\text{air}}/a)$ to $2.4(2\pi c_{\text{air}}/a)$ with the center frequency of $2.336(2\pi c_{\text{air}}/a)$. The robustness of the aforementioned frequency-doubling relationship is discussed in Figs. 10(b)–10(d) of Appendix G. In addition, Figs. 4(c), 4(f), and 4(i) show the evolution history of the optimization processes for UC2, UC3, and UC4, respectively. It is found that convergence begins within 150 iterations, and we set a maximum iteration count of 200 to obtain the final optimized result. After 200 iterations, UC2, UC3, and UC4 have all opened two bandgaps, and the center frequency of the second bandgap is twice that of the first bandgap.

As shown in Figs. 4(a), 4(d), and 4(g), the two choices of unit cell for UC2, UC3, and UC4 are indicated, similar to those of UC1. Based on the parity properties of UC2, UC3, and UC4 in Table II of Appendix C and Eq. (1), the 2D polarizations for the first bandgaps of UC2A, UC3A, and UC4B are $\mathbf{P} = (0, 0)$, indicating

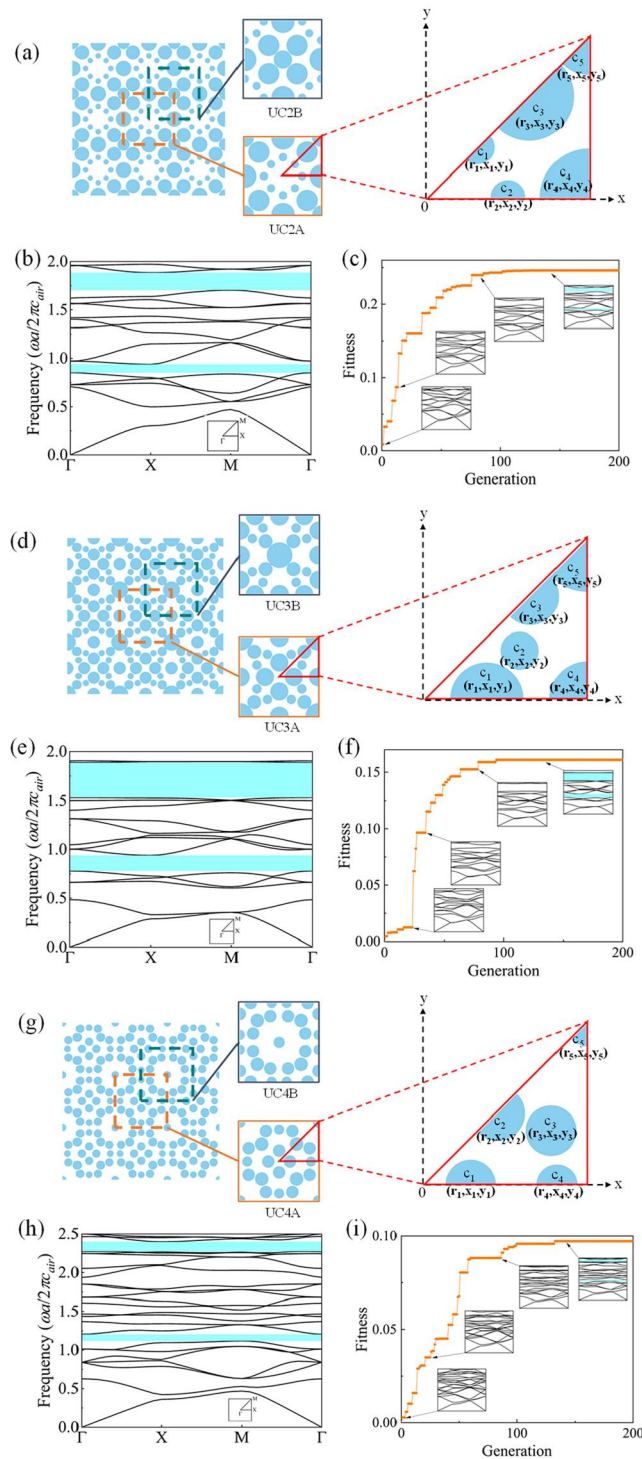


FIG. 4. Configurations of UC2 with 15 cylinders in one cell (a), UC3 with 19 cylinders in one cell (d), and UC4 with 21 cylinders in one cell (g). Band diagrams of UC2 (b), UC3 (e), and UC4 (h) after 200 iterations. Evolutionary history of GA optimization for UC2 (c), UC3 (f), and UC4 (i).

they are topologically trivial. In contrast, UC2B, UC3B, and UC4A exhibit $\mathbf{P} = (1/2, 1/2)$ for the first bandgap, indicating they are topologically nontrivial. In the same way, the second bandgaps of UC2A, UC3A, and UC4A are topologically trivial and the second bandgaps of UC2B, UC3B, and UC4B are topologically nontrivial. Therefore, the interface between UC*i*A and UC*i*B ($i = 2, 3, 4$) supports topological edge states in the two bandgaps, due to the opposite topological properties. Besides, the Q_c of UC2A, UC3A, and UC4B in the first bandgap is 0, while that of UC2B, UC3B, and UC4A is 1. Furthermore, the Q_c of UC2B, UC3B, and UC4B in the second bandgap is 1, while that of UC2A, UC3A, and UC4A is 0. UC*i*A and UC*i*B ($i = 2, 3, 4$) exhibit nonzero corner charges, indicating that topological corner states emerge within the first and second bandgaps.

In the same way with those of Fig. 2, we combine UC*i*A and UC*i*B ($i = 2, 3, 4$), and investigate the projected band diagram based on the FEM simulation, as shown in Figs. 5(a)–5(c), respectively. Topological edge states appear within the first and second bandgaps, respectively. In Fig. 5(a), the edge state along the domain wall between UC2A and UC2B in the first (second) bandgap range from $0.887(2\pi c_{\text{air}}/a)$ to $0.925(2\pi c_{\text{air}}/a)$ [range from $1.806(2\pi c_{\text{air}}/a)$ to $1.849(2\pi c_{\text{air}}/a)$]. The sound pressure fields of edge states are plotted in Fig. 5(d), at the frequencies of $0.919(2\pi c_{\text{air}}/a)$ and $1.838(2\pi c_{\text{air}}/a)$, respectively, corresponding to the frequencies indicated by dots in Fig. 5(a). In Fig. 5(b), the edge states along the domain wall between UC3A and UC3B in the first (second) bandgap range from $0.775(2\pi c_{\text{air}}/a)$ to $0.846(2\pi c_{\text{air}}/a)$ [range from $1.744(2\pi c_{\text{air}}/a)$ to $1.769(2\pi c_{\text{air}}/a)$]. The sound pressure fields of edge states are plotted in Fig. 5(e), at the frequencies of $0.789(2\pi c_{\text{air}}/a)$, $0.842(2\pi c_{\text{air}}/a)$, and $1.747(2\pi c_{\text{air}}/a)$, respectively, corresponding to the frequencies indicated by dots in Fig. 5(b). In Fig. 5(c), the edge states along the domain wall between UC4A and UC4B in the first (second) bandgap range from $1.133(2\pi c_{\text{air}}/a)$ to $1.205(2\pi c_{\text{air}}/a)$ [range from $2.35(2\pi c_{\text{air}}/a)$ to $2.356(2\pi c_{\text{air}}/a)$]. The sound pressure fields of edge states are plotted in Fig. 5(f), at the frequencies of $1.137(2\pi c_{\text{air}}/a)$ and $2.355(2\pi c_{\text{air}}/a)$, respectively, corresponding to the frequencies indicated by dots in Fig. 5(c). Furthermore, we also perform the second-stage optimization to broaden the bandwidths of the edge states, and the results can be found in Appendix E.

Finally, in the same way of Fig. 3, corner states are verified in the bending structures, which are composed of UC*i*A surrounded by UC*i*B ($i = 2, 3, 4$). The eigenvalue spectra of the FEM simulations are shown in Fig. 6; gray dots represent bulk bands, and blue (red) dots represent edge (corner) states. There are two corner states indicated by C_1 and C_2 in the first bandgap and second bandgap, respectively. Figure 6(a) shows the results for the structure composed of UC2A surrounded by UC2B. Figure 6(b) plots the sound pressure fields for edge and corner states, which correspond to $C_1[0.915(2\pi c_{\text{air}}/a)]$, $C_2[1.778(2\pi c_{\text{air}}/a)]$, $E_1[0.919(2\pi c_{\text{air}}/a)]$, and $E_2[1.829(2\pi c_{\text{air}}/a)]$ in Fig. 6(a). Figure 6(c) shows the results for the structure composed of UC3A surrounded by UC3B. Figure 6(d) plots the sound pressure fields for edge and corner states, which correspond to $C_1[0.858(2\pi c_{\text{air}}/a)]$, $C_2[1.537(2\pi c_{\text{air}}/a)]$, $E_1[0.832(2\pi c_{\text{air}}/a)]$, and $E_2[1.752(2\pi c_{\text{air}}/a)]$ in Fig. 6(c). Figure 6(e) shows the results for the structure composed of UC4A surrounded by UC4B. Figure 6(f) plots the sound pressure fields for edge and corner states, which correspond to $C_1[1.212(2\pi c_{\text{air}}/a)]$, $C_2[2.34(2\pi c_{\text{air}}/a)]$, $E_1[1.167(2\pi c_{\text{air}}/a)]$, and $E_2[2.358(2\pi c_{\text{air}}/a)]$.

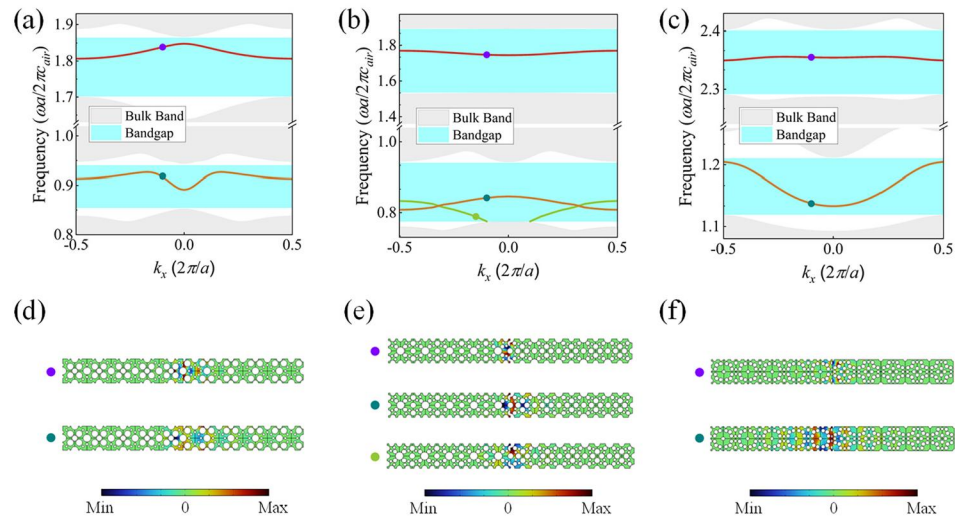


FIG. 5. Projected band diagrams of the domain wall structures composed of UC2A and UC2B (a), UC3A and UC3B (b), UC4A and UC4B (c); (d)–(f) sound pressure fields of edge states in (a)–(c).

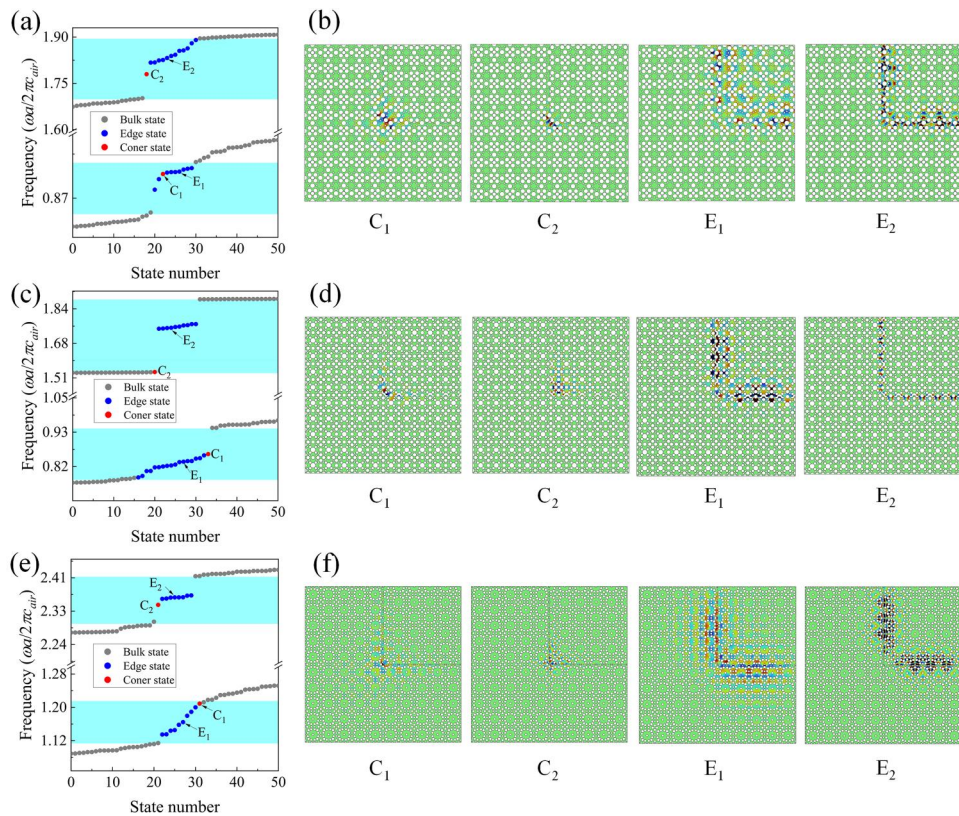


FIG. 6. Eigenvalue spectra for bending structures composed of UC2A surrounded by UC2B (a), UC3A surrounded by UC3B (c), and UC4A surrounded by UC4B (e); (b), (d), and (f) sound pressure fields for the corner states and edge states in (a), (c), and (e), respectively.

in Fig. 6(e), respectively. Obviously, the edge states are localized along the two boundaries formed by UCiA and UCiB ($i = 2, 3, 4$), and the corner states are strongly confined to the corner region. Thus, the GA optimization method is universal for designing dual-band HSTIs based on C_{4v} symmetric square lattices.

IV. CONCLUSION

By utilizing GA, dual-band STIs are achieved based on C_{4v} symmetric square lattices. Four examples, one unit cell of which contains 15, 17, 19, and 21 cylinders, respectively, are designed, and the center frequency of the second topological bandgap is twice that of the first one. By the FEM simulation, the dual-band topological edge states and double corner states are realized in the domain wall and bending structures composed of UCs with opposite topological properties, respectively. After further GA optimization, the bandwidths of edge states for both bands can be broadened. It is believed that this work provides an efficient method for designing multiband acoustic topological devices and frequency-doubled acoustic wave manipulation.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Ethics Approval

No ethical or privacy restrictions apply.

Author Contributions

Yucheng Chen: Conceptualization (equal); Data curation (equal); Formal analysis (lead); Investigation (equal); Writing – original draft (lead); Writing – review & editing (lead). **Tian Sang:** Funding acquisition (equal); Project administration (equal); Writing – review & editing (equal). **Bolun Hu:** Funding acquisition (equal); Investigation (equal); Supervision (equal); Writing – review & editing (equal). **Yueke Wang:** Conceptualization (equal); Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX A: THE PROCESS AND PSEUDOCODE OF GA

GA is a mature gradient-free optimization method, employed in this work for the dual-band optimization of STIs. We calculate fitness to achieve dual topological bandgaps and topological edge states. The population is randomly generated, which is composed

of chromosomes. Each chromosome (X_j) contains the parameter spaces (x_i , y_i , and r_i) of all cylinders within the unit cell. Therefore, the steps for GA are composed of **Initialization Parameters**, **Initialize Population**, and **Evolution Loop**, and the pseudocodes are described as follows:

Initialization Parameters:

Set maximum generations G_{max} , population size S_{pop} , crossover probability P_{cross} , mutation probability $P_{mutation}$, chromosome length L_{chrom}

Initialize Population:

For $j = 1$ to S_{pop} :

Generate random chromosome X_{j_init}

Compute fitness score $Fit_{j_init} = Fit(X_{j_init})$

Find best chromosome X_{best} with highest fitness score $Fit_{best} = \max(Fit_{j_init})$

Evolution Loop:

Current generation $G_{cur} = 0$

While current generation $G_{cur} < G_{max}$:

Generate parent chromosomes X_{parent_k} ($k = 1$ to S_{pop}) from current population based on fitness score

Randomly choose each pair of parent chromosomes, with probability P_{cross} , perform crossover to generate offspring chromosomes X_{child_m} ($m = 1$ to S_{pop})

For each X_{child_m} , with probability $P_{mutation}$, perform mutation within bounds

Perform nonlinear optimization on current X_{child_m}

Set current population $X_{j_cur} = X_{child_m}$

For $j = 1$ to S_{pop} :

Compute fitness score $Fit_{j_cur} = Fit(X_{j_cur})$

Find chromosome $X_{newbest}$ with highest fitness score $Fit_{newbest}$

if $Fit_{newbest} > Fit_{best}$:

Replace Fit_{best} with $Fit_{newbest}$

Replace X_{best} with $X_{newbest}$

Find chromosome X_{worst} with lowest fitness score

Replace X_{worst} with X_{best}

Set fitness score of X_{worst} to Fit_{best}

Generation Increment: $G_{cur} = G_{cur} + 1$

Return X_{best} as the optimal solution

where the maximum generation G_{max} , population size S_{pop} , crossover probability P_{cross} , and mutation probability $P_{mutation}$ are 200, 600, 0.6, and 0.01, respectively. The chromosome length L_{chrom} for the four UC optimization cases is set to be 8, 8, 9, and 10, respectively. The initial population is generated by random sampling within the predefined bounds for each variable. New solutions are generated through selection, crossover, and mutation operations, with mutation being implemented as a bounded random perturbation. We apply constraints to ensure all design parameters remain within their specified valid ranges. In this work, the algorithm typically converges to an optimal solution within 200 generations.

APPENDIX B: THE FITNESS SCORE FOR THE DOUBLE BANDGAP OPTIMIZATION AND PSEUDOCODE

The band diagram calculated by the FEM is stored in a matrix named B , which has M rows. M is the total number of bands. $c1$ represents probe frequency. m denotes the band order, and array B_m (B_{m+1}) denotes the $m(m+1)$ th order band. $center1$ ($center2$) is the central frequency of the first (second) bandgap.

Initialization Parameters: $\delta1_{max} = -\infty$, $\delta2_{max} = -\infty$, $center1 = 0$, $center2 = 0$

The pseudocode is as follows:

for $m = 1$ to $M-1$:
 $max_{B_m} = \max(B_m)$
 $min_{B_{m+1}} = \min(B_{m+1})$

$\delta1 = \min(c1 - max_{B_m}, min_{B_{m+1}} - c1)$
 $\delta2 = \min(2 * c1 - max_{B_m}, min_{B_{m+1}} - 2 * c1)$
 if $\delta1 > \delta1_{max}$:
 $\delta1_{max} = \delta1$
 $center1 = (min_{B_{m+1}} + max_{B_m})/2$
 if $\delta2 > \delta2_{max}$:
 $\delta2_{max} = \delta2$
 $center2 = (min_{B_{m+1}} + max_{B_m})/2$
 if $\delta1_{max} > 0$ and $\delta2_{max} > 0$:
 Both gaps are open, optimize center frequency alignment:
 $freq_{ratio} = \text{abs}(center2 - 2 * center1)$
 $fitness\ score = 1 - freq_{ratio}$
 else:
 $fitness\ score = \min(\delta1_{max}, \delta2_{max})$
 return $fitness\ score$

APPENDIX C: THE GEOMETRIC PARAMETERS OF UC($l = 1, 2, 3, 4$) AND THE PARITY OF UC/A AND UC/B ($l = 2, 3, 4$) AT POINTS OF HIGH SYMMETRY

Tables I and II.

TABLE I. Geometric parameters of UC i ($i = 1, 2, 3, 4$).

	Geometric parameters (cm)														
	c_1			c_2			c_3			c_4			c_5		
	r_1	x_1	y_1	r_2	x_2	y_2	r_3	x_3	y_3	r_4	x_4	y_4	r_5	x_5	y_5
UC1	0.260 33	0.440 66	0	0.205 25	0.599 98	0.599 98	0.199 92	1.0105	0.544 58	0.1358	1.25	1.25			
UC2	0.129 24	0.389 905	0.389 905	0.135 415	0.621 95	0	0.345 97	0.7874	0.7874	0.380 885	1.25	0	0.297 47	1.25	1.25
UC3	0.288 69	0.4622	0	0.149 32	0.719 92	0.380 95	0.222	0.8039	0.8039	0.2977	1.25	0	0.408 48	1.25	1.25
UC4	0.195 96	0.344 83	0	0.213 17	0.554 12	0.554 12	0.195 51	0.967 49	0.420 12	0.158 03	1.0132	0	0.177 23	1.25	1.25

TABLE II. Parities at high symmetry points for UC/A and UC/B ($l = 2, 3, 4$).

		Band order																				
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
UC2A	Γ	+	+	−	−	+	−	−	+	+	+	−	−	+								
	X	+	−	+	−	+	−	−	+	+	+	−	−	+								
UC2B	Γ	+	+	−	−	+	−	−	+	+	+	−	−	+								
	X	−	+	−	+	−	+	+	−	−	−	+	+	−								
UC3A	Γ	+	+	−	−	+	−	−	+	+	−	−	+	+								
	X	+	−	+	−	+	−	+	+	−	−	+	−	+								
UC3B	Γ	+	+	−	−	+	−	−	+	+	−	−	+	+								
	X	−	+	−	+	−	+	−	+	+	+	−	+	−								
UC4A	Γ	+	+	+	−	−	−	−	+	+	+	+	+	−	−	−	−	+	+	−	−	+
	X	−	+	−	−	+	+	+	−	+	+	+	−	−	−	−	+	−	+	+	−	+
UC4B	Γ	+	+	+	−	−	−	−	+	+	+	+	+	−	−	−	−	+	+	−	−	+
	X	+	−	+	+	−	−	−	+	−	−	−	+	+	+	+	−	+	−	−	+	−

APPENDIX D: OPTIMIZING THE BANDWIDTHS OF EDGE STATES FOR THE UC1 DOMAIN WALL STRUCTURE

Based on the results of Fig. 2, we further optimized the bandwidths of edge states, and the supercell was named SC1'. Considering that the energy of topological edge states is primarily localized near the interface and that the field strength decays exponentially with distance, and in order to preserve the positions and widths of the original two bandgaps, we adjusted only the two nearest-neighbor unit cells on either side of the interface. This approach allows us to most effectively modulate the bandwidth of the edge states while minimizing perturbations to the supercell band structure, reducing the number of optimization variables, and improving convergence efficiency. As shown in Fig. 7(a), only the parameters of two nearest unit cells at the interface of the

domain wall are optimized, which are named UC1A' and UC1B'. Here, we used GA to broaden the bandwidths of edge states, and the bandgaps remain consistent with those in Fig. 2. The method of evaluating fitness is described in Appendix F.

By conducting the GA, the optimized parameters of UC1A' and UC1B' are obtained in Table III, and the projected band diagrams of SC1' are shown in Fig. 7(b). Compared with the results of Fig. 2(b), the bandwidth of the edge states within the first (second) bandgap increases from $0.031(2\pi c_{\text{air}}/a)$ to $0.0526(2\pi c_{\text{air}}/a)$ [$0.0346(2\pi c_{\text{air}}/a)$]. The sound pressure fields are shown in Fig. 7(c), where the frequencies are $0.683(2\pi c_{\text{air}}/a)$ and $1.331(2\pi c_{\text{air}}/a)$, respectively, corresponding to the selected frequencies indicated by dots in Fig. 7(b). It can be seen that the sound pressure fields are highly localized around the interface between UC1A' and UC1B'. Figure 7(d) shows the evolutionary history of

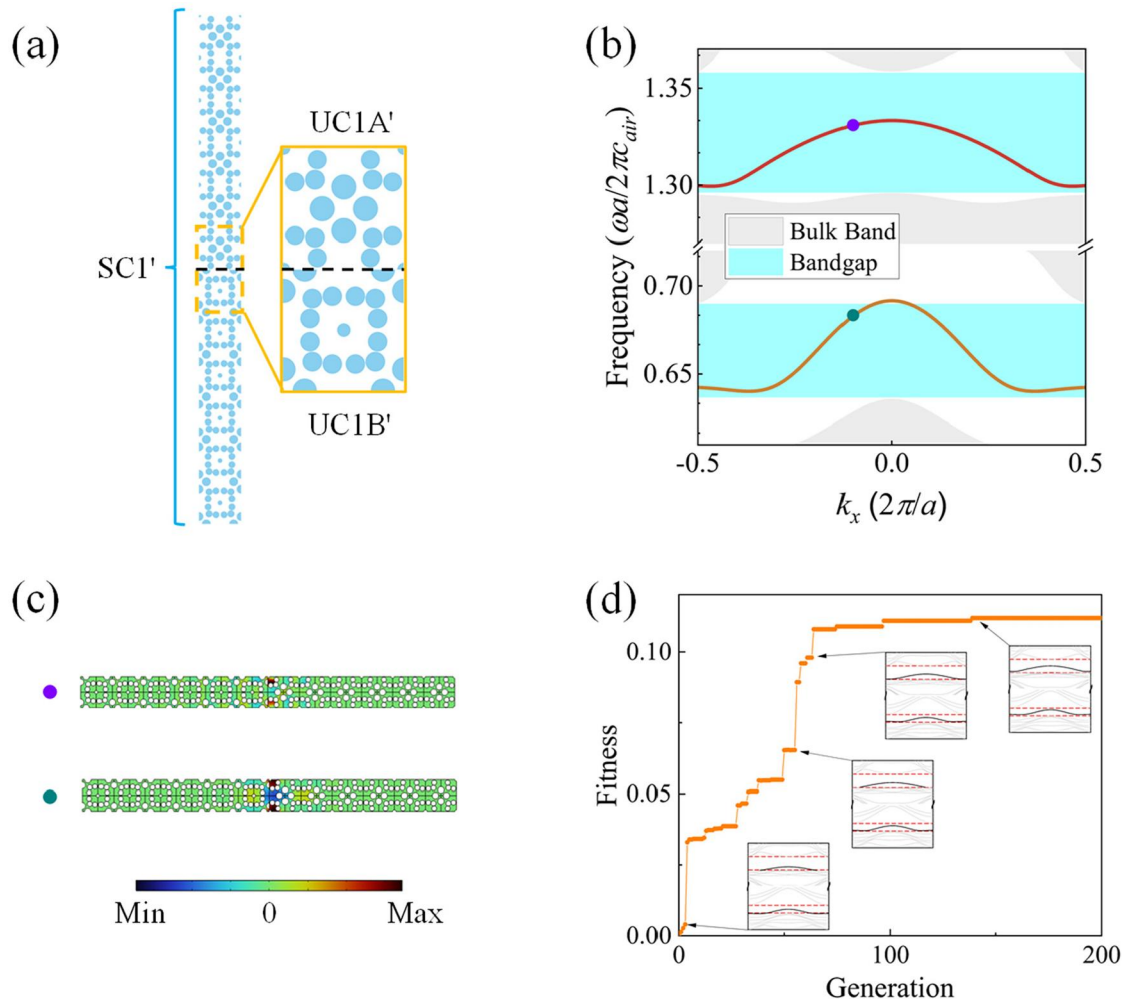


FIG. 7. (a) Supercell SC1' with the modified two nearest unit cells (UC1A' and UC1B') at the domain wall interface. (b) Projected band diagrams of (a). (c) sound pressure fields of the edge states indicated by dots in (b). (d) Evolutionary history of GA optimization for SC1'.

TABLE III. Optimized geometrical parameters for all cylinders of UC*i*A' and UC*i*B' (*i* = 1, 2, 3, 4).

	Geometric parameters (cm)														
	c_1			c_2			c_3			c_4			c_5		
	r_1	x_1	y_1	r_2	x_2	y_2	r_3	x_3	y_3	r_4	x_4	y_4	r_5	x_5	y_5
UC1A'	0.259 88	0.448 07	0	0.205 58	0.597 82	0.597 82	0.196 68	1.0102	0.550 48	0.142 15	1.25	1.25			
UC1B'	0.250 61	0.4396	0	0.204 88	0.598 09	0.598 09	0.2022	1.0109	0.549 26	0.14	1.25	1.25			
UC2A'	0.154 94	0.385	0.385	0.138 49	0.625 45	0	0.340 53	0.783 25	0.783 25	0.374 11	1.25	0	0.289 155	1.25	1.25
UC2B'	0.161 135	0.389 35	0.389 35	0.141 16	0.6319	0	0.352 495	0.782 05	0.782 05	0.373 24	1.25	0	0.291 15	1.25	1.25
UC3A'	0.283 66	0.4594	0	0.153 99	0.725 89	0.38	0.215 15	0.8	0.8	0.295 23	1.25	0	0.401 64	1.25	1.25
UC3B'	0.272 84	0.456 29	0	0.140 09	0.737 15	0.381 71	0.217 82	0.800 82	0.800 82	0.291 06	1.25	0	0.405 35	1.25	1.25
UC4A'	0.189	0.335 46	0	0.208 36	0.549 43	0.549 43	0.203 21	0.986 01	0.438 06	0.1379	1.018	0	0.1805	1.25	1.25
UC4B'	0.1855	0.3349	0	0.202 83	0.549 73	0.549 73	0.207 48	0.982 99	0.435 68	0.132 56	1.0189	0	0.186 53	1.25	1.25

the GA optimization for SC1'. The four insets in Fig. 7(d) represent the projected band diagrams of SC1' during the iteration process. The fitness value stabilizes at the 151st generation, and we set a maximum iteration count of 200 to obtain the final optimized result. Thus, the broadening of the bandwidths of edge states for the UC1 is achieved.

APPENDIX E: OPTIMIZING THE BANDWIDTHS OF EDGE STATES FOR THE DOMAIN WALL STRUCTURES BASED ON UC2, UC3, AND UC4

Following the same optimization strategy for UC1 domain wall structure, we optimized the edge states of UC2, UC3, and

UC4 domain wall structures, with the optimized supercells named SC2', SC3', and SC4', respectively. The optimized parameters for UC2A', UC3A', UC4A', UC2B', UC3B', and UC4B' are shown in Table III of Appendix D. Figures 8(a)–8(c) show the projected band diagrams for SC2', SC3', and SC4'. In Fig. 8(a), the bandwidths of the edge states within the first (second) bandgap increase from $0.04(2\pi c_{\text{air}}/a)[0.0428(2\pi c_{\text{air}}/a)]$ to $0.0402(2\pi c_{\text{air}}/a)[0.1067(2\pi c_{\text{air}}/a)]$ compared with the results of Fig. 5(a). The sound pressure fields are shown in Fig. 8(d), where the frequencies are $0.912(2\pi c_{\text{air}}/a)$, $1.806(2\pi c_{\text{air}}/a)$, and $1.852(2\pi c_{\text{air}}/a)$, respectively, corresponding to the selected frequencies indicated by dots in Fig. 8(a). In Fig. 8(b), the bandwidths of the edge states within the first (second) bandgap increase from $0.0714(2\pi c_{\text{air}}/a)[0.0252$

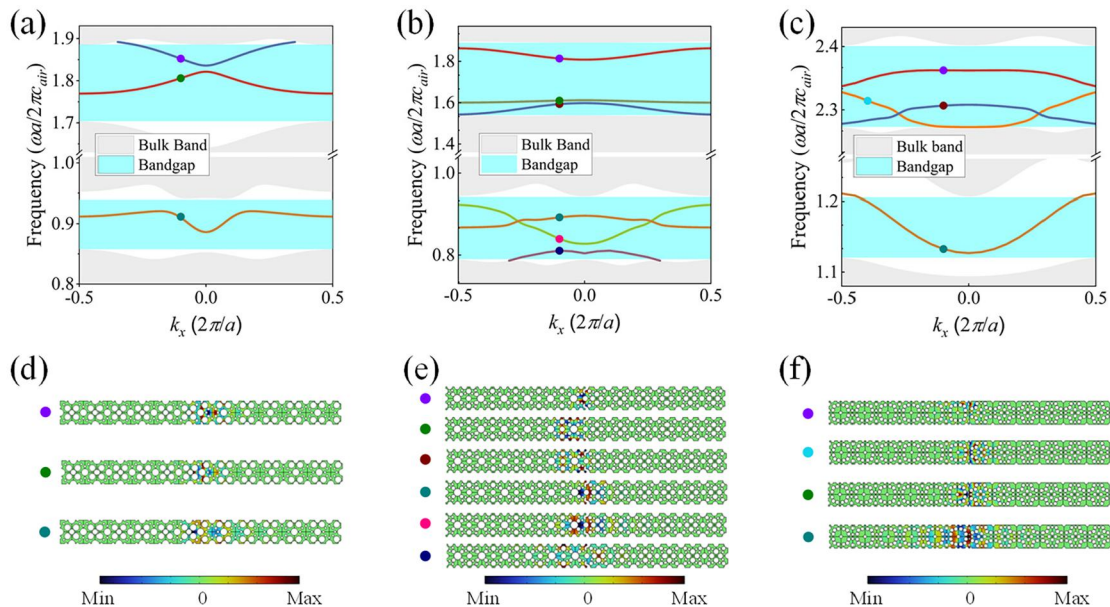


FIG. 8. Projected band diagrams of the supercell (a) SC2', (b) SC3', and (c) SC4'. Sound pressure fields (d)–(f) of edge states indicated by dots in (a)–(c), respectively.

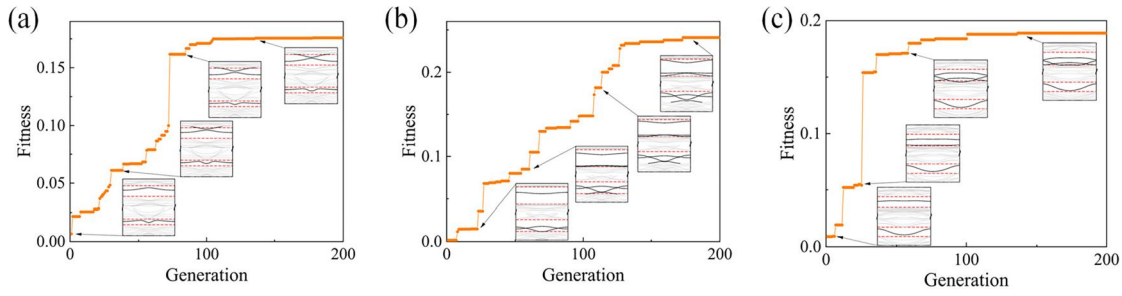


FIG. 9. Evolutionary history of GA optimization for (a) SC2', (b) SC3', and (c) SC4', respectively.

$(2\pi c_{\text{air}}/a)$ to $0.118(2\pi c_{\text{air}}/a)$ [$0.125(2\pi c_{\text{air}}/a)$] compared with the results of Fig. 5(b). The sound pressure fields are shown in Fig. 8(e), where the frequencies are $0.81(2\pi c_{\text{air}}/a)$, $0.84(2\pi c_{\text{air}}/a)$, $0.893(2\pi c_{\text{air}}/a)$, $1.594(2\pi c_{\text{air}}/a)$, $1.61(2\pi c_{\text{air}}/a)$, and $1.814(2\pi c_{\text{air}}/a)$, respectively, corresponding to the selected frequencies indicated by dots in Fig. 8(b). In Fig. 8(c), the bandwidths of the edge states within the first (second) bandgap increase from $0.0721(2\pi c_{\text{air}}/a)$ [$0.0065(2\pi c_{\text{air}}/a)$] to $0.0786(2\pi c_{\text{air}}/a)$ [$0.0799(2\pi c_{\text{air}}/a)$] compared with the results of Fig. 5(c). The sound pressure fields are shown in Fig. 8(f), where the frequencies are $1.133(2\pi c_{\text{air}}/a)$, $2.307(2\pi c_{\text{air}}/a)$, $2.314(2\pi c_{\text{air}}/a)$, and $2.362(2\pi c_{\text{air}}/a)$, respectively, corresponding to the selected frequencies indicated by dots in Fig. 8(c). These results show that the sound pressure fields are highly localized around the respective interfaces between UC2A' and UC2B', UC3A' and UC3B', as well as UC4A' and UC4B', indicating that the edge states introduced within the bandgaps remain topologically protected under C_{4v} symmetry.

Figures 9(a)–9(c) show the evolutionary history of SC2', SC3', and SC4' during optimization, respectively. The four insets in Fig. 9(a) represent the projected band diagrams of SC2' during the iteration process. Extra edge states emerge within the second bandgap at the 38th generation and fitness value stabilizes at the 138th generation, with a maximum iteration count of 200 to obtain the final optimized result. In Fig. 9(b), the four insets represent the projected band diagrams of SC3' during the iteration process. At the 60th generation, additional edge states appear in both the first and second bandgaps. The fitness value stabilizes at the 182nd generation, with a maximum iteration count of 200 to obtain the final optimized result. In Fig. 9(c), the four insets represent the projected band diagrams of SC4' during the iteration process. At the 25th generation, additional edge states emerge in the second bandgap. The fitness value stabilizes at the 145th generation, with a maximum iteration count of 200 to obtain the final optimized result. Thus, the method for broadening the bandwidths of edge states is universal.

APPENDIX F: THE FITNESS SCORE FOR WIDENING BANDWIDTH OF EDGE STATE AND PSEUDOCODE

The projected band diagrams calculated by the FEM are stored in a matrix named $\mathbf{B}$. Further, the upper and lower boundaries of the bandgap are denoted as array T_{up} and T_{down} . The edge

state is denoted as array E_i , with its bandwidth represented as BW_i . In addition to maximizing BW_i , introducing new edge states within the bandgap can also increase the bandwidth of the edge states. Consider the two bands near E_i as an example, the bands above and below E_i are array E_{i+1} and E_{i-1} , respectively. The pseudocode is as follows:

```

Calculate bandwidth of the edge state band:
 $BW_i = \max(E_i) - \min(E_i)$ 
Check if additional bands have entered the bandgap:
if  $\max(E_{i-1}) > T_{\text{down}}$  or  $\min(E_{i+1}) < T_{\text{up}}$ :
    if  $\max(E_{i-1}) > T_{\text{down}}$ :
         $\text{fitness score} = \max(E_{i-1}) - T_{\text{down}} + BW_i$ 
    else:
         $\text{fitness score} = T_{\text{up}} - \min(E_{i+1}) + BW_i$ 
else:
     $\text{fitness score} = BW_i$ 
return  $\text{fitness score}$ 

```

APPENDIX G: ROBUSTNESS ANALYSIS OF THE FREQUENCY-DOUBLING RELATION UNDER GEOMETRIC PERTURBATIONS

To evaluate the robustness of this frequency-doubling relationship under varying parameters, we conducted a systematic sensitivity analysis. While maintaining C_{4v} symmetry, we first applied random independent perturbations of $\pm 1\%$, $\pm 2\%$, and $\pm 5\%$ to the 12 geometric parameters ($r_1, x_1, y_1, r_2, x_2, y_2, r_3, x_3, y_3, r_4, x_4, y_4$) within the $1/8$ unit of UC1, generating 150 samples for each perturbation level. For each sample, we recalculated the band diagram and extracted the ratio of the center frequencies of the two bandgaps. As shown in Fig. 10(a), the results include all raw data points (blue semi-transparent scatter points), the mean $\pm$ standard deviation error bars of the valid samples, and the ideal ratio line (red dashed line). We can find that within the error range ($\pm 1\%$ to $\pm 2\%$), the ratios are very close to 2.0 (deviation $< 0.5\%$), and the standard deviation is extremely small (0.01–0.02), indicating that the frequency-doubling relationship exhibits excellent robustness; when the perturbation increases to $\pm 5\%$, the

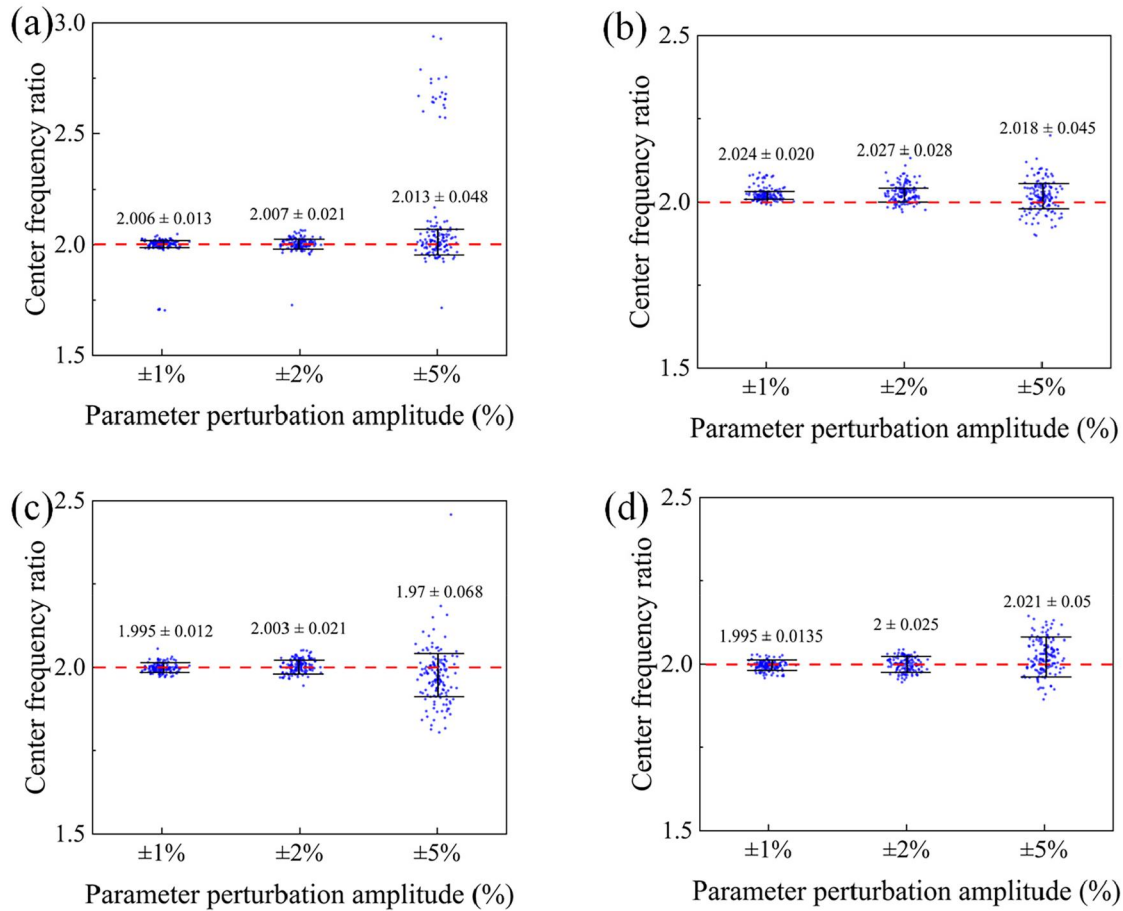


FIG. 10. Robustness of the frequency-doubling relation under different perturbations for (a) UC1, (b) UC2, (c) UC3, and (d) UC4.

standard deviation expands to 0.048, indicating that the harmonic relationship is sensitive to extreme errors. In addition, the ratios for some samples deviated significantly from 2.0. This was caused by direct contact between the epoxy resin cylinders in the samples, which closed the air channels and ultimately resulted in a sharp change in the band structure. Similar to UC1, we applied random independent perturbations of $\pm 1\%$, $\pm 2\%$, and $\pm 5\%$ to the geometric parameters of UC2, UC3, and UC4, respectively, generating 150 samples for each perturbation level while strictly maintaining C_{4v} symmetry. The results are shown in Figs. 10(b)–10(d). Within the range of $\pm 1\%$ to $\pm 2\%$, the mean deviation of the ratio of all structures from the ideal value of 2.0 is less than 0.6%, with standard deviations as low as 0.012 to 0.028, which fully demonstrates the robustness of the harmonic ratio under minor errors. When the perturbation is increased to $\pm 5\%$, the standard deviation rises to approximately 0.045–0.068. In summary, UC2, UC3, and UC4 (containing 15, 19, and 21 cylinders, respectively) all exhibit the same high robustness as UC1. Therefore, under feasible fabrication conditions, the harmonic relationship of the optimized dual-bandgap acoustic topological insulator is highly stable.

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